Fitted Q Evaluation Without Bellman Completeness via Stationary Weighting

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Abstract

Fitted Q-evaluation (FQE) is a foundational method for off-policy evaluation in reinforcement learning, but existing theory typically relies on Bellman completeness of the function class, a condition often violated in practice. We trace this reliance to a fundamental *norm mismatch*: the Bellman operator is γ-contractive in the L^2 norm induced by the target policy's stationary distribution, whereas standard FQE fits Bellman regressions under the behavior distribution. To resolve this mismatch, we reweight each Bellman regression step by an estimate of the stationary density ratio, inspired by emphatic weighting in temporal-difference learning. This makes the update behave as if it were performed under the target stationary distribution, restoring contraction without Bellman completeness while preserving the simplicity of regression-based evaluation. Illustrative experiments, including Baird's classical counterexample, show that stationary weighting can stabilize FQE under off-policy sampling.

1. Introduction

Fitted Q-evaluation (FQE), a form of approximate Bellman iteration for policy evaluation, is a standard tool in reinforcement learning (Tsitsiklis & Van Roy, 1996; Gordon, 1995; Munos, 2005; Munos & Szepesvári, 2008). Its modularity makes it particularly appealing in offline and batch RL: each iteration constructs one-step Bellman targets for policy evaluation and regresses them onto a function class, allowing practitioners to plug in off-the-shelf supervised learners (Fujimoto et al., 2019; Agarwal et al., 2021). In principle, the quality of the resulting estimator is governed by how well the function class can approximate the policy-evaluation fixed point (Farahmand et al., 2010; Agarwal et al., 2019).

Despite its practical appeal, FQE can be fragile under function approximation and distribution shift, as reflected in analyses that rely on Bellman completeness (Gordon, 1995; Baird et al., 1995; Tsitsiklis & Van Roy, 1996; Munos, 2003; Munos & Szepesvári, 2008; Melo et al., 2008; Scherrer, 2014; Fan et al., 2020; Hu et al., 2025). Bellman completeness requires that the function class be closed under the Bellman operator, so that each one-step Bellman target remains representable within the class. When this fails, projection errors can accumulate across iterations and may even grow exponentially. This instability is also reflected in recent information-theoretic hardness results for misspecified or offline fitted value and Q-evaluation (Chen & Jiang, 2019; Foster et al., 2021; Wang et al., 2021; Amortila et al., 2020). These limitations have motivated minimax and adversarial Q-learning methods that relax completeness through saddle-point formulations (Dai et al., 2018; Uehara et al., 2020; Xie et al., 2021; Uehara et al., 2021; 2023; Jin et al., 2021; Xie & Jiang, 2021), but at the cost of minimax objectives, auxiliary critic classes, and a departure from standard supervised learning with its many off-the-shelf algorithms. Related recent work has established existence of projected fixed points, and in some cases convergence to them, without Bellman completeness (Meyn, 2024; Lim & Lee, 2025; Chen et al., 2020). However, such results do not typically imply linear-rate convergence of fitted value iteration.

A norm-mismatch perspective. *The core instability of standard FQE arises from a mismatch between the regression norm and the norm in which the Bellman operator is contractive.* For policy evaluation, the Bellman operator $\mathcal{T}$ is γ-contractive in the L^2 norm induced by the target policy's stationary distribution, whereas standard FQE performs regression in the behavior (sampling) norm, where such contraction need not hold (Bertsekas & Tsitsiklis, 1996; Hallak et al., 2016; Patterson et al., 2022). Classical guarantees control this mismatch only through realizability and (approximate) Bellman completeness.

Related instability issues have long been recognized in the TD-learning literature (Tsitsiklis & Van Roy, 1996; Precup et al., 2001; Yu & Bertsekas, 2012; Mahmood et al., 2015; Patterson et al., 2022). There, TD-style updates are naturally stable under on-policy stationary sampling but can become unstable off policy. This motivated emphatic weighting (Mahmood et al., 2015), which reweights TD updates to stabilize, and in particular render contractive, the associated projected Bellman operator under off-policy sampling (Yu, 2015; Sutton et al., 2016; Hallak et al., 2016; Hallak & Mannor, 2017). Most of this literature studies TD-style methods in finite-state settings with linear function approximation, with more recent nonlinear extensions (Jiang et al., 2021; Patterson et al., 2022). Its implications for regression-based

fitted methods such as FQE, however, are less clear.

We show that FQE admits an analogous weighting-based correction adapted to the fitted setting. Because FQE proceeds by regressing Bellman targets under the behavior distribution, the appropriate remedy is to change only the regression geometry: reweight each step by stationary-density-ratio weights. This restores contraction under misspecification while preserving the simplicity of fitted Bellman iteration and avoiding minimax or saddle-point reformulations.

1.1. Our main contributions.

- We formalize *stationary-weighted* FQE for offline policy evaluation and show that the resulting projected Bellman operator is contractive without requiring Bellman completeness.
- We develop a novel finite-sample analysis for stationary-weighted FQE that separates finite-iteration, statistical, approximation, and weight-estimation errors, extending the inexact fixed-point framework of Munos & Szepesvári (2008).
- We further analyze FQE with learned rewards, such as in inverse RL, and quantify the resulting additional error. This also yields guarantees for a variant based on samples from the stationary distribution rather than explicit reweighting.

Scope and limitations. Our focus in this paper is primarily conceptual and theoretical. In particular, we bring the geometric intuition behind emphatic weighting in off-policy TD learning to regression-based fitted value methods, and show how stationary reweighting can stabilize fitted Q-evaluation. Key limitations are the need for some degree of stationary-behavior coverage and the difficulty of density-ratio estimation. Although there is a large literature on stationary density-ratio estimation (see Section 3.2) and related emphatic weights more broadly (Jiang et al., 2021; Patterson et al., 2022), obtaining stable and accurate weights under nonlinear function approximation remains challenging in practice. Our goal here is not to solve that practical problem, but rather to isolate the role of stationary reweighting as a stabilizing mechanism for regression-based fitted value methods under distribution mismatch.

1.2. Related Work

Relation to classical analyses. Classical analyses of fitted value iteration and fitted Q-evaluation typically take one of two approaches. One assumes that the Bellman operator is already contractive in the regression norm, possibly with a degraded contraction factor (Gordon, 1995; Tsitsiklis & Van Roy, 1996). The other, exemplified by Munos & Szepesvári (2008), tracks how one-step approximation and estimation errors accumulate across iterations. Such guarantees often require strong control over how the Bellman operator acts on the function class, for example through Bellman-completeness-type conditions (Chen & Jiang, 2019; Zanette et al., 2020), strong linearity assumptions (Melo & Ribeiro, 2007; Duan et al., 2020; Yang & Wang, 2020; Jin et al., 2023), or modified algorithms such as piecewise-constant partitioning methods (Peng & Williams, 1993; Xie & Jiang, 2021) and minimax Bellman-error methods (Uehara et al., 2020; Xie et al., 2021). Motivated by emphatic weighting in TD learning (Mahmood et al., 2015), we instead introduce a weighting under which the projected Bellman operator is contractive, yielding linear convergence.

Weighted temporal-difference methods. Emphatic weighting and related off-policy TD methods use reweighting to improve the stability of projected Bellman updates under distribution mismatch (Mahmood et al., 2015; Sutton et al., 2016; Hallak et al., 2016; Patterson et al., 2022), building on earlier work on weighted TD-learning (Precup et al., 2001) and weighted Bellman equations (Yu & Bertsekas, 2012; Yu et al., 2018). These works are closely related in spirit to ours: in both settings, weighting is used to recover a more favorable contraction geometry under off-policy sampling.

Our setting and analysis differ in several ways. Prior emphatic-weighting work focuses on TD-style methods, often in finite-state settings with linear function approximation (Mahmood et al., 2015; Sutton et al., 2016). With the exception of Patterson et al. (2022), these methods typically use emphatic or follow-on weights to stabilize TD recursions, rather than the stationary-density-ratio weighting we use here for fitted Bellman regression. By contrast, we study fitted value methods, where weighting enters through the Bellman *regression* step rather than a TD recursion, with a focus on nonlinear function approximation. In addition, prior analyses typically assume contraction under the weights used by the method (Patterson et al., 2022), whereas we require contraction only under the true weights and explicitly quantify how error in the estimated weights affects fitted Bellman iteration. Finally, our results clarify how weighting interacts with Bellman completeness: under Bellman completeness, the choice of weights is irrelevant, whereas outside this regime appropriate weighting can restore contraction, with approximate Bellman completeness attenuating weight error.

Minimax and pessimistic Bellman-error methods. A parallel line of work controls worst-case Bellman error using adversarial or pessimistic critics (Uehara et al., 2020; 2021; Jin et al., 2021; Xie & Jiang, 2021; Di et al., 2023; Amortila et al., 2024). These methods optimize saddle-point objectives, introduce auxiliary critic classes, and rely on assumptions such as partial coverage and dual realizability.

They are related to our approach in that the critic class often implicitly approximates discounted occupancy-ratio weights (see, e.g., Section 6 of Xie et al., 2021). Unlike these methods, however, we retain the standard regression form of FQE and modify only the weighting of the regression step.

Stationary density-ratio estimation. A large literature studies density-ratio estimation for off-policy evaluation, including stationary-distribution estimators (Glynn & Henderson, 1998; Paulin, 2015; Meggendorfer, 2023) and discounted-occupancy or stationary-discounted ratios (Nachum et al., 2019a; Zhang et al., 2020; Lee et al., 2021; 2022; Uehara et al., 2020; Nachum et al., 2019b; Wang et al., 2023; Mao et al., 2024; van der Laan et al., 2025b). DICE-style methods use such ratios to construct off-policy value estimators. Our work uses them differently: to stabilize FQE by aligning its regression geometry with the stationary norm.

2. Problem Setup

2.1. Preliminaries

We consider a discounted Markov decision process (MDP) with continuous state space $\mathcal{S}$, finite action space $\mathcal{A}$, transition kernel $P(\cdot \mid s, a)$, reward function $r_0(s, a)$, and discount factor $\gamma \in [0, 1)$. A target policy $\pi(a \mid s)$ induces the Markov operator

$$(\pi P Q)(s, a) := \mathbb{E}_\pi[Q(S', A') \mid S = s,\ A = a]\,,$$

where $S' \sim P(\cdot \mid s, a)$ and $A' \sim \pi(\cdot \mid S')$. The associated policy-evaluation Bellman operator is (Bellman, 1952; 1966)

$$\mathcal{T}Q := r_0 + \gamma\,(\pi P Q).$$

The Q-function, or state-action value function, is

$$Q^\star(s, a) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t r_0(S_t, A_t) \;\middle|\; S_0 = s,\ A_0 = a\right],$$

and is the unique fixed point of the Bellman operator, satisfying $Q^\star = \mathcal{T}Q^\star$.

We observe i.i.d. transitions $\mathcal{D}_n = \{(S_i, A_i, R_i, S_i')\}_{i=1}^n$, where $(S, A) \sim \nu_b$, $S' \sim P(\cdot \mid S, A)$, and $R = r_0(S, A) + \varepsilon$ with $E[\varepsilon \mid S, A] = 0$. Here ν_b denotes the behavior distribution on $\mathcal{S} \times \mathcal{A}$. As is standard (Xie & Jiang, 2021), we focus on i.i.d. transition sampling for simplicity; the analysis extends to dependent trajectory data under appropriate mixing conditions.

For any policy π and function f, we write

$$(\pi f)(s) := \sum_{a \in \mathcal{A}} \pi(a \mid s)\, f(s, a).$$

For any distribution ν on $\mathcal{S} \times \mathcal{A}$, let $L^2(\nu)$ denote the space of measurable functions $f : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ such that

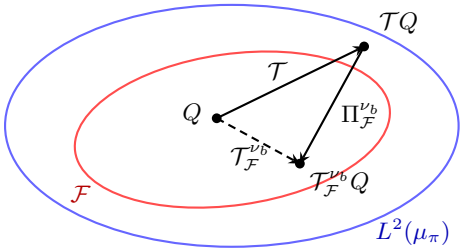

Figure 1. Norm mismatch: $Q \in \mathcal{F}$ (red), $\mathcal{T}Q$ leaves $\mathcal{F}$ but lies in $L^2(\mu_\pi)$ (blue), and $\Pi_{\mathcal{F}}^{\nu_b}$ projects it back under the behavior-norm $L^2(\nu_b)$. The projection $\mathcal{T}_{\mathcal{F}}^{\nu_b} = \Pi_{\mathcal{F}}^{\nu_b}\mathcal{T}$ need not be contractive.

$\int f(s, a)^2\, \nu(ds, da) < \infty$, equipped with norm

$$\|f\|_{2,\nu} := \left(\int_{\mathcal{S}\times\mathcal{A}} f(s, a)^2\, \nu(ds, da)\right)^{1/2}.$$

Stationary distributions. A distribution μ_π on $\mathcal{S} \times \mathcal{A}$ is *stationary* for (π, P) if it is preserved by one-step evolution under policy π and transition kernel P, that is, $\mu_\pi = \mu_\pi \pi P$. Intuitively, for an ergodic Markov chain induced by (π, P), μ_π is the long-run state-action distribution (Puterman, 1990).

2.2. Standard FQE and Norm Mismatch

Classical dynamic programming computes the Q-function $Q^\star$ by iterating the Bellman operator $\mathcal{T}$, viewing value estimation as a Picard iteration toward the fixed point $Q^\star = \mathcal{T}Q^\star$ (Bellman, 1966). In tabular settings this can be done exactly, but in larger or continuous state spaces one must combine Bellman updates with function approximation over a class $\mathcal{F}$ (Sutton et al., 1998). Standard fitted Q-evaluation (FQE) does so by constructing iterates

$$\widehat{Q}^{(k+1)} \in \arg\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \big(R_i + \gamma\,(\pi \widehat{Q}^{(k)})(S_i') - f(S_i, A_i)\big)^2,$$

that is, by repeatedly fitting least-squares regressions to one-step Bellman targets under the behavior distribution ν_b. In the population limit, this corresponds to iterating the projected Bellman operator $\mathcal{T}_{\mathcal{F}}^{\nu_b} := \Pi_{\mathcal{F}}^{\nu_b}\mathcal{T}$, where $\Pi_{\mathcal{F}}^{\nu_b}$ denotes the $L^2(\nu_b)$ projection onto $\mathcal{F}$.

Classical analyses (Munos & Szepesvári, 2008) effectively assume that iterating $\mathcal{T}_{\mathcal{F}}^{\nu_b}$ closely tracks iteration of $\mathcal{T}$ in an appropriate norm, so that iterating $\mathcal{T}_{\mathcal{F}}^{\nu_b}$ converges to the unique fixed point $Q^\star$ of $\mathcal{T}$ up to an inherent Bellman error. A standard sufficient condition is *Bellman completeness* of $\mathcal{F}$, meaning $\mathcal{T}f \in \mathcal{F}$ for all $f \in \mathcal{F}$; in this case, $\Pi_{\mathcal{F}}^{\nu_b}\mathcal{T}f = \mathcal{T}f$ on $\mathcal{F}$, so the projection does not interfere with the contraction of $\mathcal{T}$ (Chen & Jiang, 2019).

When $\mathcal{F}$ is not Bellman complete, $\mathcal{T}_{\mathcal{F}}^{\nu_b}$ need not be contractive. The issue is geometric: $\Pi_{\mathcal{F}}^{\nu_b}$ is nonexpansive in $L^2(\nu_b)$,

whereas $\mathcal{T}$ is often analyzed as contracting in a different norm, such as $\|\cdot\|_\infty$. These norms need not be comparable (Figure 1). As a result, standard FQE may fail to contract, and even when it converges, the limiting behavior-projected solution can be far from the true fixed point $Q^\star$ (Patterson et al., 2022).

2.3. On-Policy Stability and Resolving Norm Mismatch

A natural way to resolve this mismatch is to ask which norm gives the right geometry for Bellman iteration. A classical result points to the answer: under on-policy sampling from the stationary distribution μ_π of the target policy, Bellman updates are stable under function approximation because the Bellman operator is contractive and projection is nonexpansive in $L^2(\mu_\pi)$ (Bertsekas & Tsitsiklis, 1996; Hallak et al., 2016; Patterson et al., 2022). In that setting, the fitted update projects Bellman targets in $L^2(\mu_\pi)$ rather than in the behavior norm $L^2(\nu_b)$, so the projected Bellman operator is also contractive. This in turn suggests a simple remedy for off-policy FQE: reweight the regression step so that it behaves as if the data were drawn from μ_π, echoing the motivation behind emphatic weighting in TD learning.

To formalize this, let $\mathcal{F} \subseteq L^2(\mu_\pi)$ be convex, and let $\overline{\mathcal{F}}$ denote its closure in $L^2(\mu_\pi)$. We define the projected Bellman operator in $L^2(\mu_\pi)$ by

$$\mathcal{T}_{\mathcal{F}} := \Pi_{\mathcal{F}}\mathcal{T}, \qquad \Pi_{\mathcal{F}} g := \arg\min_{f\in\overline{\mathcal{F}}} \|f-g\|_{2,\mu_\pi}. \tag{1}$$

C1 Existence of stationary distribution. The MDP induced by (π, P) admits a stationary distribution μ_π.

C2 Convex function class. $\mathcal{F}$ is a convex subset of $L^2(\mu_\pi)$.

Lemma 1 (Contraction of Bellman operator)**.** *Assume C1. Then the Bellman operator $\mathcal{T}$ is a γ-contraction on $L^2(\mu_\pi)$:*

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_{2,\mu_\pi} \le \gamma \|Q - Q'\|_{2,\mu_\pi}, \qquad \forall\, Q, Q' \in L^2(\mu_\pi).$$

If C2 also holds, then the projected operator $\mathcal{T}_{\mathcal{F}}$ is a γ-contraction on $\mathcal{F}$.

Condition C1 holds under mild ergodicity conditions (Puterman, 1990). Condition C2 is standard and ensures that $\Pi_{\mathcal{F}}$ is well defined and nonexpansive (Munos & Szepesvári, 2008; Patterson et al., 2022). It is satisfied by many common classes, including linear and RKHS-based function classes.

3. Stationary-Weighted Fitted Q-Evaluation

3.1. Algorithm

Motivated by the on-policy stability of FQE, we replace the behavior-weighted regression loss in standard FQE with a weighted loss targeting the stationary $L^2(\mu_\pi)$ geometry. As

Algorithm 1 Stationary-Weighted Fitted Q-Evaluation

input Offline data $\mathcal{D}_n$, target policy π, function class $\mathcal{F}$, iterations K, discount γ

```
# Estimate stationary density ratio
```

$\widehat{d}_\pi \leftarrow$ estimator of d_π

Initialize $\widehat{Q}^{(0)} \in \mathcal{F}$

for $k = 0, \dots, K-1$ **do**

```
  # Construct targets and weights
```

  $Y_i^{(k)} \leftarrow R_i + \gamma\,(\pi\widehat{Q}^{(k)})(S_i')$

  $w_i \leftarrow \widehat{d}_\pi(S_i, A_i)$

```
  # Stationary-weighted regression
```

  $\widehat{Q}^{(k+1)} \leftarrow \arg\min_{f\in\mathcal{F}} \sum_{i=1}^n w_i \big\{Y_i^{(k)} - f(S_i, A_i)\big\}^2$

end for

output $\widehat{Q}^{(K)}$

we show in Section 4.2, our theory only requires an *admissible* weighting function, of which the stationary density ratio is one natural choice.

Define the stationary density ratio for (π, P) by

$$d_\pi(s,a) := \frac{d\mu_\pi}{d\nu_b}(s,a),$$

assuming $\mu_\pi \ll \nu_b$. Given an estimate $\widehat{d}_\pi$, let $w_i := \widehat{d}_\pi(S_i, A_i)$. We then define the *stationary-weighted* FQE update by

$$\widehat{Q}^{(k+1)} = \arg\min_{f\in\mathcal{F}} \frac{1}{n}\sum_{i=1}^n w_i \Big\{R_i + \gamma(\pi\widehat{Q}^{(k)})(S_i') - f(S_i, A_i)\Big\}^2.$$

When $\widehat{d}_\pi = d_\pi$, this update corresponds to projection in $L^2(\mu_\pi)$.

Algorithm 1 summarizes the procedure, and Section 4 provides finite-sample guarantees. The approach is compatible with generic function approximators, including neural networks and boosted ensembles. Its computational cost is essentially the same as that of standard FQE: after a one-time density-ratio estimation step, it performs the same K supervised regressions over $\mathcal{F}$.

Interpretation of the stationary norm. Beyond being the norm in which contraction holds, $L^2(\mu_\pi)$ is a natural geometry for policy evaluation because it weights prediction error by the target policy's long-run visitation frequencies. Since μ_π depends only on (π, P), this norm is invariant to the behavior policy and the initial-state distribution, and thus reflects long-run prediction performance rather than the particulars of the data-collection process (Puterman, 2014; Levin & Peres, 2017).

3.2. Estimation of Stationary Ratios

A key component of stationary-weighted FQE is estimating the stationary density ratio. This is challenging in its

own right and typically leads to a minimax or saddle-point problem, as in DICE-style methods (Nachum et al., 2019a; Zhang et al., 2020; Lee et al., 2021; Wang et al., 2023) and related minimax weight-estimation approaches (Uehara et al., 2020). Stationary ratios are closely related to the discounted occupancy ratios widely used in RL, and arise as their limit as $\gamma \to 1$ under standard ergodicity conditions.

Since $d_\pi \nu_b = \mu_\pi$ is a stationary distribution for (π, P), the stationary ratio $d_\pi = d\mu_\pi / d\nu_b$ satisfies the moment condition

$$\mathbb{E}\big[d_\pi(S,A)\,\big(g(S,A) - g(S',A')\big)\big] = 0 \quad \text{for all } g, \quad (2)$$

where $(S,A) \sim \nu_b$, $S' \sim P(\cdot \mid S,A)$, and $A' \sim \pi(\cdot \mid S')$.

Minimax and DICE-style saddle-point methods enforce stationarity through a variational formulation and learn the ratio using paired weight and critic classes. A representative empirical objective is

$$\arg\min_{d \in \mathcal{D}} \sup_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^{n} d(S_i, A_i)\,\big(g(S_i,A_i) - g(S_i', A_i')\big) - \frac{1}{2n} \sum_{i=1}^{n} g(S_i, A_i)^2 \right\} + \left(\frac{1}{n} \sum_{i=1}^{n} d(S_i, A_i) - 1 \right)^2,$$

where $A_i' \sim \pi(\cdot \mid S_i')$, and the final term enforces normalization. When the critic class $\mathcal{G}$ is linear or an RKHS, the inner maximization can often be expressed in closed form (Olivas-Martinez & Rotnitzky, 2025). In practice, such structured critic classes can yield simpler and more stable estimators than highly flexible saddle-point methods, while still producing useful weights; see, for example, approximate balancing weights (Wang et al., 2023). Theoretical analyses of minimax weight estimators are given by Dikkala et al. (2020); Uehara et al. (2020); Wang et al. (2023).

Importantly, weighting can help even without accurate stationary-ratio estimation. Because the weights define only the regression norm, even misspecified weights may improve stability by better aligning the update with the on-policy geometry. See Patterson et al. (2022) and Section 4.4 for further discussion.

4. Theoretical Guarantees

We study the ERM version of stationary-weighted FQE in Algorithm 1. Extensions to more general function approximators follow from standard sample-splitting arguments (Munos & Szepesvári, 2008; Farahmand et al., 2010). For nonconvex classes, one may instead analyze contraction on a suitable convex hull and control the induced approximation error separately.

4.1. Projected Bellman Fixed Point

The population target of stationary-weighted FQE is the fixed point $Q^\star_{\mathcal{F}}$ of the projected Bellman operator

$$\mathcal{T}_{\mathcal{F}} := \Pi_{\mathcal{F}} \mathcal{T}.$$

If $Q^\star \in \mathcal{F}$, then $Q^\star_{\mathcal{F}} = Q^\star$. More generally, when realizability fails, the next result shows that the gap between $Q^\star_{\mathcal{F}}$ and $Q^\star$ is controlled by the approximation error of $Q^\star$ in $\mathcal{F}$.

Theorem 1 (Function approximation error)**.** *Under C1–C2,*

$$\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \;\le\; \frac{1}{1-\gamma} \inf_{f \in \mathcal{F}} \|f - Q^\star\|_{2,\mu_\pi}.$$

Thus, $Q^\star_{\mathcal{F}}$ lies within a factor $(1-\gamma)^{-1}$ of the best $L^2(\mu_\pi)$ approximation to $Q^\star$ in $\mathcal{F}$.

Our analysis then proceeds in three steps: (i) identify stationary-weighted FQE as an inexact γ-contractive Picard iteration toward $Q^\star_{\mathcal{F}}$; (ii) control the resulting Bellman approximation and finite-iteration errors; and (iii) bound sampling and density-ratio estimation error.

4.2. Admissible Weighting Functions

Our theory does not require the exact stationary density ratio. Instead, it only assumes the existence of a weighting function that induces the same projection geometry on $\mathcal{F}$ as the stationary norm. This requirement is strictly weaker than Bellman completeness, $\mathcal{T}(\mathcal{F}) \subseteq \mathcal{F}$, under which the projection geometry is irrelevant and any weighting function is admissible.

C3 Bellman residual orthogonality under weighting. There exists nonnegative $d_\pi \in L^2(\nu_b)$ such that, for all $f, Q \in \mathcal{F}$,

$$\mathbb{E}_{\nu_b}\big[d_\pi(S,A) f(S,A) \{\mathcal{T}Q - \Pi_{\mathcal{F}}\mathcal{T}Q\}(S,A)\big] = 0.$$

Condition C3 is the first-order optimality condition ensuring that the projected update

$$\mathcal{T}_{\mathcal{F}}(Q^{(k)}) = \arg\min_{f \in \mathcal{F}} \|\mathcal{T}(Q^{(k)}) - f\|_{2,\mu_\pi}$$

can equivalently be computed by d_π-weighted regression under the behavior distribution ν_b:

$$\arg\min_{f \in \mathcal{F}} \mathbb{E}_{\nu_b}\Big[d_\pi(S,A)\{\mathcal{T}(Q^{(k)})(S,A) - f(S,A)\}^2\Big].$$

Equivalently, Condition C3 requires the Bellman residual $\mathcal{T}Q - \Pi_{\mathcal{F}}\mathcal{T}Q$ to be orthogonal to $\mathcal{F}$ in $L^2(d_\pi \nu_b)$. Under Bellman completeness, this holds automatically because the residual is zero. When d_π is taken to be the stationary

density ratio, C3 amounts to an overlap requirement between the behavior and stationary distributions.

Hereafter, let d_π denote any weighting function satisfying Condition C3, and let $\hat{d}_\pi$ denote an estimator of it. A natural choice is the stationary density ratio, though other weighting schemes, such as those motivated by emphatic TD learning, may also be adapted to our setting (Patterson et al., 2022).

Relation to dual realizability in minimax Q-learning. Condition C3 is related to, but distinct from, the assumptions used in minimax, saddle-point, and related moment-based approaches to Q-function estimation (Uehara et al., 2020; Xie et al., 2021; Uehara et al., 2021; 2023; Zanette & Wainwright, 2022). Those methods typically require both coverage of the target discounted occupancy measure and dual realizability of the relevant ratio or moment equation within a critic class (Section 6 of Xie et al. (2021)). The former is closely related to C3 in long-horizon settings, since the discounted occupancy ratio converges to the stationary ratio d_π as $\gamma \to 1$ under standard ergodicity conditions (Meyn & Tweedie, 1993). By contrast, our approach does not require dual realizability within a critic class, only sufficiently accurate estimation of an admissible weight.

4.3. Error Propagation Under Bellman Iteration

We first control how per-iteration regression errors accumulate over the iterative update. The following lemma adapts the standard deterministic inexact Picard iteration bound of Munos & Szepesvári (2008) to the projected Bellman operator $\mathcal{T}_\mathcal{F}$ in the stationary $L^2(\mu_\pi)$ geometry.

Lemma 2 (Approximate Picard iteration error bound)**.** *Assume C1–C2. If $\{\eta_k\}_{k\geq 1}$ satisfies*

$$\|\hat{Q}^{(k)} - \mathcal{T}_\mathcal{F}\hat{Q}^{(k-1)}\|_{2,\mu_\pi} \;\leq\; \eta_k \qquad \textit{for all } k,$$

then for any $K \geq 1$,

$$\|\hat{Q}^{(K)} - Q^\star_\mathcal{F}\|_{2,\mu_\pi} \;\leq\; \gamma^K \|\hat{Q}^{(0)} - Q^\star_\mathcal{F}\|_{2,\mu_\pi} + \sum_{j=1}^{K} \gamma^{K-j}\, \eta_j.$$

Per-iteration regression error. We use the above lemma with inexactness terms η_k defined as high-probability bounds on the regression errors $\|\widehat{Q}^{(k)} - \mathcal{T}_\mathcal{F}\widehat{Q}^{(k-1)}\|_{2,\mu_\pi}$.Unlike Munos & Szepesvári (2008), our setting requires controlling weight-estimation error in addition to the usual sampling error.

We measure regression complexity via the extended class

$$\mathcal{F}_{\text{ext}} := \mathcal{F} \cup \{\, \pi f : f \in \mathcal{F} \,\}.$$

For $\varepsilon > 0$, let $N(\varepsilon, \mathcal{F}_{\text{ext}}, L^2(Q))$ denote the ε-covering number of $\mathcal{F}_{\text{ext}}$ under $L^2(Q)$, and define the uniform entropy integral (Van Der Vaart & Wellner, 1996)

$$\mathcal{J}(\delta, \mathcal{F}_{\text{ext}}) := \int_0^\delta \sup_Q \sqrt{\log N(\varepsilon, \mathcal{F}_{\text{ext}}, L^2(Q))}\, d\varepsilon,$$

where the supremum is over discrete distributions Q. The statistical accuracy of each regression step is then governed by the critical radius

$$\delta_n := \inf\Big\{\delta > 0 : \frac{\mathcal{J}(\delta, \mathcal{F}_{\text{ext}})}{\sqrt{n}\,\delta^2} \leq 1\Big\}.$$

We impose the following mild technical conditions, several of which are standard and can be relaxed with more refined analysis.

C4 Uniform boundedness. The quantities $\hat{d}_\pi$, d_π, d_π^{-1}, R, and $\{\hat{Q}^{(k)}, \mathcal{T}_\mathcal{F}\hat{Q}^{(k)}\}_{k=0}^K$ are uniformly bounded by a constant $M < \infty$.

C5 Stationary–weighted norm equivalence. There exist $0 < c_0 \leq c_1 < \infty$ such that for all $f \in L^2(\mu_\pi)$,

$$c_0\|f\|_{2,\mu_\pi} \;\leq\; \|f\|_{2,\, d_\pi \nu_b} \;\leq\; c_1\|f\|_{2,\mu_\pi}.$$

C6 Sample splitting. The estimator $\hat{d}_\pi$ is obtained from data independent of $\mathcal{D}_n$.

C7 Complexity growth condition. The entropy integral of the extended class satisfies $\mathcal{J}(\infty, \mathcal{F}_{\text{ext}}) < \infty$ and $\mathcal{J}(\delta, \mathcal{F}_{\text{ext}})/\{\delta\sqrt{\text{loglog}(1/\delta)}\} \to \infty$ as $\delta \to 0$.

Condition C4 can be relaxed using weight clipping and appropriate tail conditions on the density ratio and outcomes, following variance-control ideas from emphatic- and importance-weighted RL (Hallak et al., 2016; Han & Sung, 2019; Mehrabi & Wager, 2024) and heavy-tailed regression. Condition C6 can similarly be relaxed by standard cross-fitting arguments (Nie & Wager, 2021; Foster & Syrgkanis, 2023). Condition C5 holds automatically under Condition C4 when d_π is the stationary density ratio. More generally, it is an overlap condition requiring that the weighted behavior measure $d_\pi\nu_b$ provide sufficient coverage of μ_π to compare the corresponding L^2-norms. Condition C7 controls localized Rademacher complexity (Bartlett et al., 2005; Munos & Szepesvári, 2008).

We now present our **main technical lemma**. It depends on the *inherent Bellman error* (Munos & Szepesvári, 2008)

$$\varepsilon_{\text{Bell}} := \sup_{Q\in\mathcal{F}} \|\mathcal{T}(Q) - \mathcal{T}_\mathcal{F}(Q)\|_{2,\mu_\pi},$$

which quantifies the deviation from Bellman completeness of $\mathcal{F}$ in the stationary norm. To obtain sharper rates under approximate Bellman completeness, we assume that the

function class $\mathcal{F}$ satisfies a Hölder-type relation between the supremum norm and the stationary $L^2(\mu_\pi)$ norm. This condition holds with $\alpha = 0$ for bounded classes and with $\alpha > 0$ for many common structured classes.

C8 Hölder-type sup–L^2 relation. There exist constants $\alpha \in [0,1]$ and $L < \infty$ such that, for all $k \in [K]$, almost surely,

$$\|\mathcal{T}\hat{Q}^{(k)} - \hat{Q}^{(k+1)}\|_\infty \;\le\; L\,\|\mathcal{T}\hat{Q}^{(k)} - \hat{Q}^{(k+1)}\|_{2,\mu_\pi}^{\alpha}.$$

For finite-dimensional linear models of dimension p, this holds with $L \asymp \sqrt{p}$ and $\alpha = 1$. For reproducing kernel Hilbert spaces with eigenvalue decay $\lambda_j \asymp j^{-2r}$, one may take $\alpha = \frac{2r}{2r+1}$ (Mendelson & Neeman, 2010, Lemma 5.1). It also holds for signed convex hulls of suitable bases (van de Geer, 2014, Lemma 2), and for d-variate Hölder and Sobolev classes of order $s > d/2$ on bounded domains, where $\alpha = 1 - \frac{d}{2s}$ (Bibaut et al., 2021, Lemma 4) (Adams & Fournier, 2003; Triebel, 2006).

Lemma 3 (One-step weighted regression error bound)**.** *Assume C1-C8. There exists a constant $C = C(M, L) > 0$ such that, for all $\eta \in (0,1)$, with probability at least $1-\eta$,*

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k)}) - \hat{Q}^{(k+1)}\|_{2,\mu_\pi} \le\; C\Big(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}}\Big) + C\Big(\Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi}^{\frac{1}{1-\alpha}} + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi}\varepsilon_{\mathrm{Bell}}^{\alpha}\Big).$$

When $\alpha = 0$, we have $1/(1-\alpha) = 1$, so the final two terms reduce to $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_{2,\mu_\pi}$. For $\alpha > 0$, the bound is sharper when the inherent Bellman error $\varepsilon_{\mathrm{Bell}}$ is small, that is, under approximate Bellman completeness. Thus, stronger structure leads to less sensitivity of the one-step error to weight-estimation error. If one instead assumes supremum-norm control of the weight error, then C8 is not needed, and the final two terms can be replaced, up to constants, by $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_\infty\,\varepsilon_{\mathrm{Bell}}$; see Lemma 10 in the appendix.

For s-sparse linear classes in ambient dimension p, $\delta_n \asymp \sqrt{s\log(p/s)/n}$ (Wainwright, 2019). For VC-subgraph dimension V, $\delta_n \asymp \sqrt{V\log(n/V)/n}$ (Van Der Vaart & Wellner, 1996). For Hölder/Sobolev classes of smoothness s with dimension $d \le 2s$, $\delta_n \asymp n^{-s/(2s+d)}$ (Nickl & Pötscher, 2007; Mendelson & Neeman, 2010). For bounded Hardy–Krause variation, $\delta_n \asymp n^{-1/3}(\log n)^{2(d-1)/3}$ (Bibaut & van der Laan, 2019).

4.4. Main Finite-Sample Guarantee

Our main result follows by combining the high-probability regression error bound in Lemma 3 with the deterministic inexact Picard iteration bound in Lemma 2.

Theorem 2 (Finite-Sample Error Bound for Stationary-Weighted FQE)**.** *Assume C1–C8. There exists a constant $C = C(M, L) > 0$ such that, for all $\eta \in (0,1)$ and $K \in \mathbb{N}$, with probability at least $1-\eta$,*

$$\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \le \gamma^K\,\|\hat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} + \frac{C}{1-\gamma}\Big(\delta_n + \sqrt{\tfrac{\log(K/\eta)}{n}}\Big) + \frac{C}{1-\gamma}\Big(\Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi}^{\frac{1}{1-\alpha}} + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi}\varepsilon_{\mathrm{Bell}}^{\alpha}\Big).$$

Theorem 2 parallels the classical sup-norm bounds of Munos & Szepesvári (2008), but in the stationary $L^2(\mu_\pi)$ norm and without Bellman completeness. A key difference is the additional density-ratio error term, which in the worst case scales as $\|\hat{d}_\pi/d_\pi - 1\|_{2,\mu_\pi}/(1-\gamma)$ (corresponding to $\alpha = 0$, after absorbing $\varepsilon_{\mathrm{Bell}}$ into constants). The bound is most useful in the consistent regime, where $\|\hat{d}_\pi/d_\pi - 1\|_{2,\mu_\pi} \to 0$. For inconsistent estimators, one can instead analyze weighted FQE under the misspecified weights, possibly with an inflated contraction factor. Theorem 1 further shows that the approximation error of $Q^\star_{\mathcal{F}}$ is bounded, up to a $(1-\gamma)^{-1}$ factor, by the best approximation error of $Q^\star$ in $\mathcal{F}$, namely $\inf_{f\in\mathcal{F}}\|Q^\star - f\|_{2,\mu_\pi}$.

The generic weight-estimation term in Theorem 2 can be made more explicit using guarantees for particular minimax or saddle-point estimators. For example, analyses of minimax estimation in conditional moment models similar to (2) are given by Dikkala et al. (2020); Olivas-Martinez & Rotnitzky (2025), while related minimax estimators for off-policy weighting are studied by Uehara et al. (2020); Wang et al. (2023). At a high level, these results suggest that the weight-estimation term decomposes into

$$\underbrace{\inf_{d\in\mathcal{D}}\|d - d^\dagger\|}_{\text{weight-class approximation}} + \underbrace{\inf_{g\in\mathcal{G}}\|g - g^\dagger\|}_{\text{critic-class approximation}} + \underbrace{\mathrm{stat}(\mathcal{D},\mathcal{G},n)}_{\text{statistical error}},$$

where $\mathcal{D}$ and $\mathcal{G}$ are the weight and critic classes, $d^\dagger$ is the target weighting function, and the statistical term is controlled by the localized complexity of these classes; see also Section 5 of Bennett et al. (2023). Thus, the corresponding term in our bound inherits the usual combination of weight approximation, critic approximation, and statistical error.

Robustness to weight error under approximate Bellman completeness. Theorem 2 shows that small inherent Bellman error $\varepsilon_{\mathrm{Bell}}$ attenuates the effect of weight misspecification in stationary-weighted FQE. In particular, the weight-error contribution is

$$\Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi}^{\frac{1}{1-\alpha}} + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,\mu_\pi}\varepsilon_{\mathrm{Bell}}^{\alpha}.$$

When $\left\|\frac{\hat{d}_\pi}{d_\pi} - 1\right\|_{2,\mu_\pi} < 1$, the first term is smaller than linear in the weight error whenever $\alpha > 0$, while the second is damped by $\varepsilon_{\text{Bell}}^{\alpha}$. As $\alpha \to 1$, the contribution is approximately $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_{2,\mu_\pi}\, \varepsilon_{\text{Bell}}$, so the effect of weight error is small when either the weights are accurate or the inherent Bellman error is small.

Contraction under approximate weights. Our analysis extends beyond exact recovery of the stationary density ratio or other admissible weights. It is enough that the learned weights induce a norm under which $\mathcal{T}$ is contractive, possibly with a larger contraction factor $\overline{\gamma} \in (0, 1)$ in place of γ. Thus, even if $\hat{d}_\pi$ converges to a misspecified limit $\overline{d}_\pi$, the resulting weighted norm $L^2(\nu_b \overline{d}_\pi)$ may still improve on the behavior norm used by standard FQE whenever $\mathcal{T}$ contracts in that norm, echoing the motivation behind emphatic weighting in TD learning (Patterson et al., 2022).

A simple sufficient condition is that $\overline{d}_\pi$ be multiplicatively close to the stationary density ratio d_π:

$$(1 - \varepsilon)\, d_\pi(s, a) \;\le\; \overline{d}_\pi(s, a) \;\le\; (1 + \varepsilon)\, d_\pi(s, a).$$

Then $L^2(\nu_b \overline{d}_\pi)$ and $L^2(\nu_b d_\pi)$ are equivalent, and $\mathcal{T}$ remains a $\overline{\gamma}$-contraction in $L^2(\nu_b \overline{d}_\pi)$, with $\overline{\gamma} \le \gamma \sqrt{\frac{1+\varepsilon}{1-\varepsilon}}$. Thus, small multiplicative errors preserve contraction up to a modest inflation of the contraction factor.

On-policy stability and ergodic limits. Setting $\hat{d}_\pi \equiv 1$ recovers the unweighted FQE update. When the data are drawn on-policy from μ_π, the regression norm is exactly $L^2(\mu_\pi)$, so the update contracts toward $Q^\star_{\mathcal{F}}$. More generally, off-policy degradation is controlled by $\|d_\pi^{-1} - 1\|_{2,\mu_\pi}$. Under ergodicity and on-policy trajectory sampling, empirical visitation frequencies converge to μ_π, so the norm mismatch vanishes asymptotically and unweighted FQE inherits the same contraction geometry.

4.5. Stationary Sampling with Estimated Rewards

In some simulator-based environments, one can sample directly from the stationary distribution μ_π of the target policy. In this case, stationary reweighting is unnecessary, and the main error source shifts from density-ratio estimation to reward estimation. This setting is relevant when rewards must be learned from data, as in robotics and inverse reinforcement learning (Abbeel & Ng, 2004; Kober et al., 2013; Fu et al., 2018; van der Laan et al., 2025c;a).

Concretely, suppose P—and hence μ_π—is known. Then we may sample $(\tilde{S}_i, \tilde{A}_i, \tilde{S}_i') \sim \mu_\pi \otimes P$ directly and avoid stationary weighting altogether. Let $\hat{r}(s, a)$ be a reward estimate learned from the behavior data $\mathcal{D}_n$. Running FQE on $\{(\tilde{S}_i, \tilde{A}_i, \hat{r}(\tilde{S}_i, \tilde{A}_i), \tilde{S}_i')\}_{i=1}^n$ is equivalent to applying Algorithm 1 with $d_\pi \equiv 1$. We analyze the resulting error in $\hat{Q}^{(K)}$ due to reward misspecification.

Define the reward-estimated Bellman operator

$$\hat{\mathcal{T}}_{\mathcal{F}} := \Pi_{\mathcal{F}} \hat{\mathcal{T}}, \qquad \hat{\mathcal{T}} Q := \hat{r} + \gamma\, \pi P Q.$$

Let $\hat{Q}^\star_{\mathcal{F}}$ denote the fixed point of $\hat{\mathcal{T}}_{\mathcal{F}}$, which is the target of the estimator $\hat{Q}^{(K)}$.

By the triangle inequality,

$$\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \;\le\; \|\hat{Q}^{(K)} - \hat{Q}^\star_{\mathcal{F}}\|_{2,\mu_\pi} + \|\hat{Q}^\star_{\mathcal{F}} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi}.$$

The first term is controlled directly by Theorem 2. The second term captures the effect of reward misspecification; the next theorem bounds this perturbation.

Theorem 3 (Reward Misspecification)**.** *Under C1–C2,*

$$\|\hat{Q}^\star_{\mathcal{F}} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \;\le\; \frac{1}{1-\gamma}\, \|\hat{r} - r\|_{2,\mu_\pi}.$$

Moreover, if $\mathcal{F}$ is a linear subspace of $L^2(\mu_\pi)$, then

$$\|\hat{Q}^\star_{\mathcal{F}} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \;\le\; \frac{1}{1-\gamma}\, \|\Pi_{\mathcal{F}}(\hat{r} - r)\|_{2,\mu_\pi}.$$

For linear classes, only the component $\Pi_{\mathcal{F}}(\hat{r} - r)$ of the reward error affects the fixed point. Thus, one may estimate r within the same class used for FQE, provided stationary-weighting is used, without requiring reward realizability.

5. Illustrative Experiments

We illustrate the theory using two controlled experiments designed to isolate the effect of norm mismatch beyond Bellman completeness. The first uses Baird's classical counterexample to study slowed contraction under limited overlap between the stationary and behavior distributions (Baird et al., 1995). The second uses 100-state random sparse Dirichlet (Garnet-style) MDPs to illustrate exponential divergence under severe norm mismatch (Bhatnagar et al., 2009).

Experiment 1: Stationary weighting restores geometric decay. We consider the 7-state "hub-and-spokes" MDP from Baird et al. (1995) (Appendix A), a canonical counterexample for instability in approximate dynamic programming. The target policy always selects the *solid* action, which deterministically moves the agent to the hub, while the behavior policy selects it with probability $\kappa \in (0, 1]$ (otherwise taking the *dashed* action, which moves uniformly to a spoke), so smaller κ corresponds to weaker stationary overlap. We use the standard 7-dimensional linear feature map and run linear FQE, comparing unweighted regression to stationary-weighted regression using the *exact* stationary cdensity ratio. We vary κ from 1 to 0.7 with $\gamma = 0.95$. Errors are reported in the behavior norm $\|Q^{(K)} - Q^\star\|_{2,\nu_b}$ under the empirical behavior distribution.

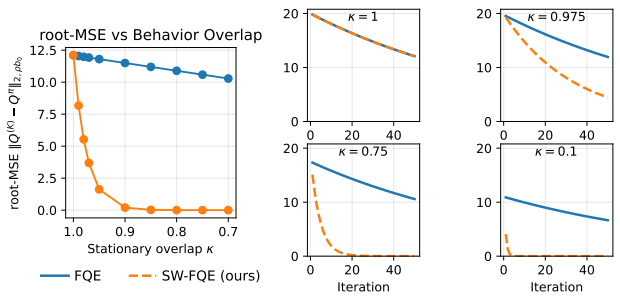

Figure 2. Moderate-overlap regime ($\gamma = 0.95$): root mean squared error (RMSE) at iteration 100 versus κ (left), and iteration curves showing RMSE versus iteration number at various κ (right).

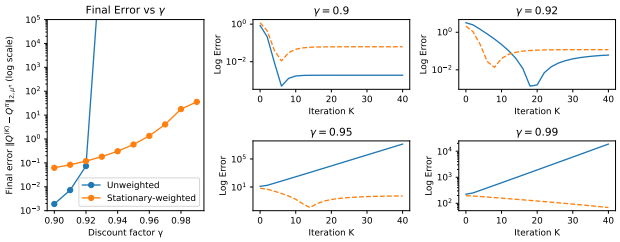

Figure 3. Severe norm-mismatch regime: final error vs. γ (left) and iteration curves (right).

Figure 2 shows that as overlap decreases, the behavior-norm projection increasingly distorts the Bellman update and weakens its contraction: standard FQE slows dramatically, while SW-FQE maintains nearly uniform geometric decay across all κ. When $\kappa = 1$, the behavior and stationary distributions coincide and the two methods are identical. Results for $\gamma \in \{0.99, 0.999\}$ are deferred to Appendix A.

Experiment 2: Exponential divergence under norm mismatch. We evaluate SW-FQE on random Garnet MDPs with sparse Dirichlet transitions, using $|\mathcal{S}| = 100$ states, $|\mathcal{A}| = 4$ actions, and a fixed branching factor of 5, meaning that for each state–action pair only 5 successor states have nonzero transition probability. The target policy is sampled uniformly at random, while the behavior policy is ε-greedy with $\varepsilon = 0.1$. To amplify norm mismatch, we generate off-policy data using a `reset` sampling scheme that breaks stationarity. We use a low-dimensional linear function class that contains $Q^\star$ exactly but is not Bellman-complete, and compare unweighted FQE to stationary-weighted FQE using the exact stationary density ratios. We sweep $\gamma \in \{0.90, 0.925, 0.95, 0.96, 0.97, 0.98, 0.99\}$, run $K = 200$ fitted iterations, and average over $M = 200$ random seeds. Full details are in Appendix A.

Figure 3 reports the final mean-squared error in the stationary norm after K iterations as a function of γ (left) and the corresponding error trajectories versus iteration for representative discounts (right). It reveals a sharp phase transition in the behavior of standard FQE as γ increases. For moderate discounts, both methods converge, with an initial decrease in error followed by a shallow *variance elbow* as iterations increase. SW-FQE can exhibit slightly larger error because stationary weighting inflates variance by upweighting rare state–action pairs under limited action overlap, whereas unweighted FQE pools and extrapolates. Beyond $\gamma \approx 0.92$, unweighted FQE becomes unstable and exhibits clear exponential divergence, despite the fact that $Q^\star$ lies exactly in the function class. In contrast, SW-FQE remains stable across the entire sweep and continues to exhibit geometric decay in the stationary norm.

Together, these experiments show that standard FQE can remain convergent yet arbitrarily slow under inadequate stationary overlap (Baird), while under severe norm mismatch it undergoes a sharp phase transition and diverges exponentially (Garnet). In both regimes, stationary weighting restores contraction-like behavior and fast geometric decay.

6. Conclusion

This work shows that weighting can stabilize fitted Bellman methods by changing the geometry of each regression update. In standard FVI/FQE, regressions are fit under the behavior distribution even though Bellman contraction is governed by the target policy's stationary distribution. Reweighting the loss toward that stationary distribution aligns the update with the norm in which the Bellman operator contracts, restoring a contraction-based analysis and linear convergence beyond Bellman completeness while preserving the simplicity of classical regression-based FVI/FQE under general function approximation.

This perspective connects fitted value methods to the emphatic-weighting literature in TD learning, where weighting has long been used to address off-policy instability. Our contribution is to show that the same principle extends beyond TD recursions and linear settings: weighting can be built directly into regression-based FVI/FQE, providing a simple mechanism for stabilizing fitted Bellman updates under misspecification and off-policy sampling. More broadly, this suggests that weighting is a useful design principle for fitted value methods, not only for TD learning.

Important practical challenges remain. Estimating stationary ratios can be difficult, and saddle-point estimators such as DICE may be unstable in practice, especially when stationary coverage is weak. An important direction for future work is therefore to develop weighting schemes that are computable, stable, and effective at improving contraction in fitted Bellman updates. Simpler weight classes, such as linear or RKHS-based classes, may offer a more stable alternative to highly flexible saddle-point estimators while still improving over the unweighted baseline.

Impact Statement.

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Experiments

A.1. Design and additional experiments (Experiment 1)

Experimental setup. We use a modified Baird-style 7-state MDP (Baird et al., 1995). There is a hub state 0 and six spokes $1, \ldots, 6$. The *solid* action deterministically transitions to state 0 from any state, while the *dashed* action transitions uniformly among the spokes. The target policy always chooses the solid action, giving a stationary distribution μ_π that places all its mass on the hub. The behavior policy chooses the solid action with probability $\kappa \in (0, 1]$ and the dashed action otherwise; smaller κ yields weaker stationary overlap because the hub is rarely visited. We use the standard 7-dimensional feature representation from Baird et al. (1995): each spoke has its own basis coordinate, and the hub's feature vector is the sum of all spoke features (plus a bias term). We generate data by simulating transitions under the behavior policy to obtain samples (S_i, A_i, R_i, S_i'), with rewards set to zero so that $Q^\star \equiv 0$. At each FQE iteration we solve a regularized linear least-squares problem in this feature space. For stationary-weighted FQE we use the exact stationary density ratio $d_\pi(a, s) \propto \mu_\pi(a, s)/\nu_b(a, s)$, which is known analytically for this MDP and is nonzero only at the hub, so the experiment isolates the effect of the norm mismatch rather than density-ratio estimation error. All reported metrics are averaged over 100 independent replicates, and errors are computed in the behavior norm $\|\cdot\|_{2,\nu_b}$ using the empirical behavior distribution from each simulated dataset.

A.1.1. Performance under the stationary norm

Figure 4 reports stationary-norm errors for the setting of Experiment 1. As in the main text, decreasing overlap weakens the effective contraction of standard FQE, leading to slow convergence, while SW-FQE continues to exhibit fast geometric decay across all values of κ. Because the stationary norm places all mass on the hub, these curves highlight the dominant error component that unweighted FQE is unable to correct under low overlap.

A.1.2. Larger discount factors

To assess sensitivity to the discount factor, we repeat the stationary-overlap experiment for $\gamma = 0.99$ and $\gamma = 0.999$. As γ increases, the Bellman operator becomes less contractive in the stationary norm, amplifying the effect of norm mismatch. Standard FQE slows substantially in these high-discount regimes, whereas SW-FQE continues to converge rapidly even when overlap is extremely small. These re-

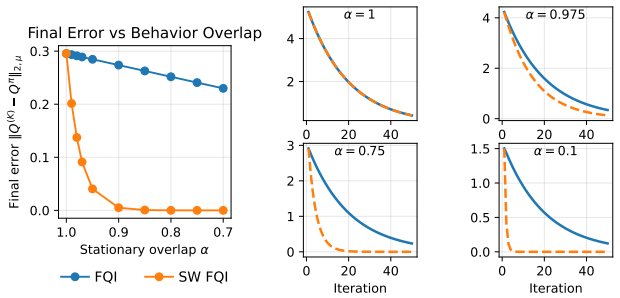

Figure 4. Final stationary-norm error versus stationary overlap for $\gamma = 0.95$. Left: $\|Q^{(K)} - Q^\star\|_{2,\mu_\pi}$ across κ. Right: iteration curves at selected overlap levels.

sults confirm that the qualitative behavior observed in the main text persists—and becomes more pronounced—as γ approaches 1.

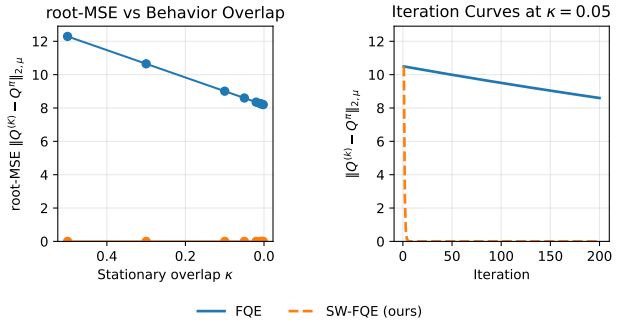

Figure 5. High-discount, low-overlap regime ($\gamma = 0.999$). Left: final error vs. κ. Right: iteration curves at $\kappa = 0.05$.

A.2. Additional details on experiment 2

A.2.1. Experimental design

We evaluate SW-FQE on random Garnet MDPs with $|\mathcal{S}| = 100$, $|\mathcal{A}| = 4$, and branching factor 5. For each state–action pair, 5 distinct next states are sampled uniformly without replacement and assigned transition probabilities via a Dirichlet distribution. Rewards are drawn i.i.d. from $\mathcal{N}(0, 1)$.

For each MDP instance, the target policy π is sampled independently for each state from a Dirichlet distribution over actions. The ground-truth value functions $(V^\pi, Q^\star)$ are computed exactly by solving the Bellman equations via linear system inversion. The stationary distribution d_π is computed by power iteration on the induced Markov chain (Bertsekas & Tsitsiklis, 1996).

The behavior policy μ_π is chosen to be ε-greedy with respect to $Q^\star$ with $\varepsilon = 0.1$, ensuring persistent but limited mismatch with the target. Off-policy datasets of size $n = 10{,}000$ are generated using the `reset` sampling scheme, in which each transition starts from an independent state drawn uniformly from $\mathcal{S}$. This deliberately breaks stationarity and amplifies the norm mismatch between the sampling distribution and the target stationary distribution.

We use a linear Q-function class of dimension $d = 5$ constructed so that $Q^\star$ lies exactly in the span of the features. Specifically, $Q^\star$ is embedded as one column of a random low-dimensional basis over $\mathcal{S} \times \mathcal{A}$, ensuring that the model is *correct* but generically *not Bellman-complete*. Thus, $\mathcal{TF} \not\subseteq \mathcal{F}$ with probability one.

Unweighted FQE is implemented via standard least-squares regression. Stationary-weighted FQE reweights each regression step using the exact stationary density ratio $w(s, a) = \frac{d_\pi(s)\pi(a|s)}{\hat{\rho}_S(s)\mu_\pi(a|s)}$, where $\hat{\rho}_S$ is the empirical state marginal induced by the dataset. Both methods use ridge regularization with $\lambda = 10^{-6}$ and are initialized at $\theta_0 = 0$.

We sweep the discount factor $\gamma \in \{0.90, 0.925, 0.95, 0.96, 0.97, 0.98, 0.99\}$ and run $K = 200$ fitted iterations. All learning curves and final errors are averaged over $M = 50$ independent random seeds for the MDP, policies, and dataset generation. Errors are reported in the stationary norm $\|Q^{(K)} - Q^\star\|_{2,\mu_{\pi^\pi}}$.

A.2.2. Additional experiments

In this experiment, stationary weighting uses *estimated* stationary density ratios obtained by solving the corresponding DualDICE min–max saddle-point problem in closed form on the tabular MDP. This introduces additional variance from ratio estimation, which can slightly weaken early-iteration performance. Nevertheless, as seen in Fig. 6, the qualitative stabilization effect persists: even with estimated weights, stationary reweighting prevents exponential divergence and preserves geometric decay under severe norm mismatch.

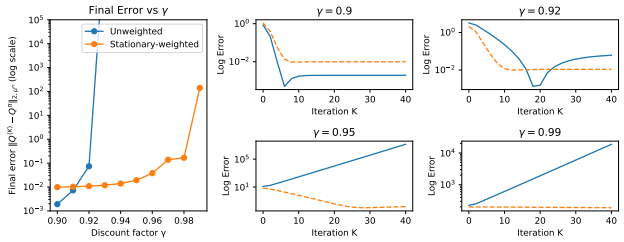

Figure 6. Severe norm-mismatch regime with *estimated* stationary density ratios: final error versus γ (left) and iteration curves (right).

B. Fixed-Point Estimation of Stationary Density Ratios

The stationary ratio $d_\pi = d\mu_\pi / d\nu_b$ is identified by the fixed-point equation implied by stationarity $\mu_\pi = \mu_\pi P_\pi$:

$$d_\pi(s', a') = (\mathcal{K} d_\pi)(s', a'),$$

where the linear operator $\mathcal{K}$ is

$$(\mathcal{K}g)(s',a') = \frac{\pi(a' \mid s')}{\nu_b(a' \mid s')}\, \mathbb{E}_{(S,A)\sim\nu_b}[g(S,A)\, k(S,A,s')]\,,$$

and

$$k(s,a,s') = \frac{P(s' \mid s,a)}{\rho_{b,1}(s')}, \rho_{b,1}(s') := \mathbb{E}_{(S,A)\sim\nu_b}[P(s' \mid S,A)]\,.$$

is the transition density ratio, with $\rho_{b,1}$ the marginal density of S' induced by the behavior dynamics. The quantity k can be estimated directly using the empirical squared-loss objective of Hines & Miles (2025):

$$\frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} k(S_i,A_i,S'_j)^2 \;-\; \frac{2}{n}\sum_{i=1}^{n} k(S_i,A_i,S'_i).$$

However, $\mathcal{K}$ is the adjoint of πP in $L^2(\mu_\pi)$ and satisfies $\|\mathcal{K}\|_{L^2(\mu_\pi)} = 1$, so direct Picard iteration of $d_\pi = \mathcal{K} d_\pi$ is not contractive. A standard workaround is the discounted resolvent equation

$$d_{\gamma'} = (1-\gamma')\tfrac{\pi}{\nu_b} + \gamma'\,\mathcal{K} w_{\gamma'},$$

whose solution $d_{\gamma'}$ is the γ'-discounted occupancy ratio. The resolvent replaces exact stationarity by a γ'-discounted flow balance, which yields a γ'-contractive update. Since $\mathcal{K}$ is nonexpansive in $L^2(\mu_\pi)$, Picard iteration applies and $w_{\gamma'}$ can be computed analogously to fitted value iteration.

Under standard ergodicity assumptions—irreducibility, aperiodicity, and geometric mixing—the discounted ratios satisfy $w_{\gamma'} \to d_\pi$ as $\gamma' \uparrow 1$ (Meyn & Tweedie, 1993; Glynn & Meyn, 1996). The resolvent formulation induces a familiar bias–variance trade-off: smaller $1-\gamma'$ improves numerical stability but introduces bias of order $O(1-\gamma')$, whereas letting $\gamma' \uparrow 1$ removes the bias at the expense of variance scaling like $(1-\gamma')^{-1}$ due to the effective horizon of the discounted occupancy measure.

C. Contraction and Error Propagation Results

C.1. Contraction Under Stationary Measures

Lemma 4 (Bellman contraction for Q-functions under a stationary measure)**.** *If μ_π is stationary for πP, then πP is nonexpansive in $L^2(\mu_\pi)$:*

$$\|\pi P(Q_1 - Q_2)\|_{2,\mu_\pi} \;\le\; \|Q_1 - Q_2\|_{2,\mu_\pi}.$$

Therefore the Bellman operator $\mathcal{T}$ is a γ-contraction:

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \;\le\; \gamma\,\|Q_1 - Q_2\|_{2,\mu_\pi}.$$

Proof. Let $h := Q_1 - Q_2$. For any (s,a), Jensen's inequality gives

$$\big(\pi P h\big)(s,a)^2 \;\le\; \big(\pi P(h^2)\big)(s,a),$$

since πP is a conditional expectation operator.

Integrating with respect to μ_π and using stationarity $\mu_\pi = \mu_\pi(\pi P)$,

$$\|\pi P h\|_{2,\mu_\pi}^2 = \int (\pi P h)^2\, d\mu_\pi \le \int \pi P(h^2)\, d\mu_\pi = \int h^2\, d\mu_\pi = \|h\|_{2,\mu_\pi}^2.$$

Thus πP is nonexpansive: $\|\pi P h\|_{2,\mu_\pi} \le \|h\|_{2,\mu_\pi}$.

For the Bellman operator,

$$\mathcal{T}Q_1 - \mathcal{T}Q_2 = \gamma\,\pi P(Q_1 - Q_2) = \gamma\,\pi P h.$$

Taking norms and applying nonexpansiveness proves

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \le \gamma \|h\|_{2,\mu_\pi},$$

as claimed. □

Lemma 5 (Contraction of the projected Bellman operator)**.** *If μ_π is stationary for πP, then $\mathcal{T}_\mathcal{F}$ is a γ-contraction on $L^2(\mu_\pi)$; that is, for all $Q_1, Q_2 \in L^2(\mu_\pi)$*

$$\|\mathcal{T}_\mathcal{F} Q_1 - \mathcal{T}_\mathcal{F} Q_2\|_{2,\mu_\pi} \;\le\; \gamma\,\|Q_1 - Q_2\|_{2,\mu_\pi}.$$

Proof. By Lemma 4,

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \le \gamma\,\|Q_1 - Q_2\|_{2,\mu_\pi}.$$

Since $\Pi_\mathcal{F}$ is the $L^2(\mu_\pi)$ orthogonal projection onto a closed convex set, it is nonexpansive:

$$\|\Pi_\mathcal{F} f - \Pi_\mathcal{F} g\|_{2,\mu_\pi} \le \|f - g\|_{2,\mu_\pi}.$$

Thus

$$\begin{aligned}\|\mathcal{T}_\mathcal{F} Q_1 - \mathcal{T}_\mathcal{F} Q_2\|_{2,\mu_\pi} &= \|\Pi_\mathcal{F}(\mathcal{T}Q_1) - \Pi_\mathcal{F}(\mathcal{T}Q_2)\|_{2,\mu_\pi} \\ &\le \|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_{2,\mu_\pi} \\ &\le \gamma\,\|Q_1 - Q_2\|_{2,\mu_\pi}.\end{aligned}$$

□

Proof of Lemma 1. The first claim follows directly from Lemma 4. The second claim follows from Lemma 5. □

C.2. Misspecification error bound

Proof of Lemma 1. Write

$$Q^\star_\mathcal{F} - Q^\star = \mathcal{T}_\mathcal{F} Q^\star_\mathcal{F} - \mathcal{T}Q^\star = \big(\mathcal{T}_\mathcal{F} Q^\star_\mathcal{F} - \mathcal{T}_\mathcal{F} Q^\star\big) + \big(\mathcal{T}_\mathcal{F} Q^\star - \mathcal{T}Q^\star\big).$$

Taking $L^2(\mu_\pi)$ norms and applying the triangle inequality

$$\|Q^\star_\mathcal{F} - Q^\star\|_{2,\mu_\pi} \le \|\mathcal{T}_\mathcal{F} Q^\star_\mathcal{F} - \mathcal{T}_\mathcal{F} Q^\star\|_{2,\mu_\pi} + \|\mathcal{T}_\mathcal{F} Q^\star - \mathcal{T}Q^\star\|_{2,\mu_\pi}.$$

By the contraction property of Lemma 5,

$$\|\mathcal{T}_\mathcal{F} Q^\star_\mathcal{F} - \mathcal{T}_\mathcal{F} Q^\star\|_{2,\mu_\pi} \le \gamma\,\|Q^\star_\mathcal{F} - Q^\star\|_{2,\mu_\pi},$$

and by definition of $\mathcal{T}_{\mathcal{F}}$,

$$\mathcal{T}_{\mathcal{F}}Q^\star - \mathcal{T}Q^\star = \Pi_{\mathcal{F}}\mathcal{T}Q^\star - \mathcal{T}Q^\star = -(I - \Pi_{\mathcal{F}})\,\mathcal{T}Q^\star.$$

Combining these two displays yields

$$\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \le \gamma\,\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} + \left\|(I - \Pi_{\mathcal{F}})\,\mathcal{T}Q^\star\right\|_{2,\mu_\pi}.$$

Rearranging, we obtain

$$(1-\gamma)\,\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \le \left\|(I - \Pi_{\mathcal{F}})\,\mathcal{T}Q^\star\right\|_{2,\mu_\pi},$$

which proves the first inequality:

$$\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \le \frac{1}{1-\gamma}\left\|(I - \Pi_{\mathcal{F}})\,\mathcal{T}Q^\star\right\|_{2,\mu_\pi}.$$

For the second inequality, note that $\Pi_{\mathcal{F}}\mathcal{T}Q^\star$ is the best $L^2(\mu_\pi)$ approximation to $\mathcal{T}Q^\star$ in $\mathcal{F}$, so

$$\left\|(I - \Pi_{\mathcal{F}})\,\mathcal{T}Q^\star\right\|_{2,\mu_\pi} = \inf_{f\in\mathcal{F}} \|\mathcal{T}Q^\star - f\|_{2,\mu_\pi}.$$

Since $Q^\star$ is a fixed point of $\mathcal{T}$, we have $\mathcal{T}Q^\star = Q^\star$, and hence

$$\left\|(I - \Pi_{\mathcal{F}})\,\mathcal{T}Q^\star\right\|_{2,\mu_\pi} = \inf_{f\in\mathcal{F}} \|Q^\star - f\|_{2,\mu_\pi}.$$

Substituting this into the previous display yields

$$\|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu_\pi} \le \frac{1}{1-\gamma}\inf_{f\in\mathcal{F}} \|f - Q^\star\|_{2,\mu_\pi},$$

which completes the proof. □

C.3. Approximate Q Iteration and Error Propagation

We now prove the error–propagation bound for approximate Q iteration with $\mathcal{T}_{\mathcal{F}}$ under the stationary measure. It is the natural analogue of the classical perturbed Picard iteration inequality for inexact fixed-point iterations of a contraction mapping.

Proof of Lemma 2. By Lemma 5, $\mathcal{T}_{\mathcal{F}}$ is a γ-contraction on $L^2(\mu_\pi)$. Let

$$a_k \;:=\; \|\hat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi}, \qquad k \ge 0,$$

with $\hat{Q}^{(0)} = \hat{Q}$. For any $k \ge 1$,

$$\begin{aligned} a_k &= \|\hat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \\ &\le \|\hat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)})\|_{2,\mu_\pi} + \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)}) - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \\ &\le \eta_k + \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)}) - \mathcal{T}_{\mathcal{F}}(Q^\star_{\mathcal{F}})\|_{2,\mu_\pi} \\ &\le \eta_k + \gamma\,\|\hat{Q}^{(k-1)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} = \eta_k + \gamma\, a_{k-1}, \end{aligned}$$

where we used the assumed approximation bound in the first term, the fixed-point property $\mathcal{T}_{\mathcal{F}}Q^\star_{\mathcal{F}} = Q^\star_{\mathcal{F}}$ in the second line, and the contraction property in the third.

Thus the sequence $\{a_k\}$ satisfies the recursion

$$a_k \le \gamma\, a_{k-1} + \eta_k, \qquad k \ge 1.$$

Unrolling this recursion yields

$$a_K \le \gamma^K a_0 + \sum_{j=1}^{K} \gamma^{K-j}\eta_j \;=\; \gamma^K\|\hat{Q} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} + \sum_{j=1}^{K}\gamma^{K-j}\eta_j,$$

which is the desired bound. □

D. Fitted Q Evaluation with Stationary Weighting

D.1. Local maximal inequality

Let $O_1, \ldots, O_n \in \mathcal{O}$ be independent random variables. For any function $f : \mathcal{O} \to \mathbb{R}$, define

$$\|f\| := \sqrt{\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[f(O_i)^2]}. \tag{3}$$

For a star-shaped class of functions $\mathcal{F}$ and a radius $\delta \in (0,\infty)$, define the localized Rademacher complexity

$$\mathcal{R}_n(\mathcal{F},\delta) := \mathbb{E}\left[\sup_{\substack{f\in\mathcal{F}\\ \|f\|\le\delta}} \frac{1}{n}\sum_{i=1}^{n}\epsilon_i f(O_i)\right],$$

where ϵ_i are i.i.d. Rademacher random variables.

The following lemma provides a local maximal inequality and restates Lemma 11 of (Foster & Syrgkanis, 2023) (see also Lemma 11 of van der Laan et al. (2025d)).

Lemma 6 (Local maximal inequality)**.** *Let $\mathcal{F}$ be a star-shaped class of functions satisfying $\sup_{f\in\mathcal{F}}\|f\|_\infty \le M$. Let $\delta = \delta_n \in (0,1)$ satisfy the critical radius condition $\mathcal{R}_n(\mathcal{F},\delta) \le \delta^2$, and suppose that, as $n \to \infty$,*

$$\frac{1}{\sqrt{n}}\sqrt{\log\log(1/\delta_n)} = o(\delta_n).$$

Then there exists a constant $C > 0$ such that, for all $\eta \in (0,1)$, with probability at least $1-\eta$, every $f \in \mathcal{F}$ satisfies

$$\begin{aligned} \frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) &\le C\Big(\delta^2 + \delta\,\|f\|\Big) \\ &+\; C\Big(\frac{\sqrt{\log(1/\eta)}\,\|f\|}{\sqrt{n}} + \frac{M\,\log(1/\eta)}{n}\Big). \end{aligned}$$

Proof. Lemma 11 in (van der Laan et al., 2025d) shows that there exists a constant $C > 0$ (possibly depending on

the choice of sequence $\{\delta_n\}$) such that, for all $u \geq 1$, with probability at least $1 - e^{-u^2}$, every $f \in \mathcal{F}$ satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i)-\mathbb{E}[f(O_i)]\big) \leq C\Big(\delta_n^2+\delta_n\,\|f\|+\frac{u\,\|f\|}{\sqrt{n}}+\frac{M\,u^2}{n}\Big).$$

Set $u := \sqrt{\log(1/\eta)}$, so that $e^{-u^2} = e^{-\log(1/\eta)} = \eta$. Substituting this choice of u into the above inequality yields, with probability at least $1-\eta$, every $f \in \mathcal{F}$ satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) \leq C\Big(\delta_n^2 + \delta_n\,\|f\|\Big) + C\Big(\frac{\sqrt{\log(1/\eta)}\,\|f\|}{\sqrt{n}} + \frac{M\,\log(1/\eta)}{n}\Big).$$

□

The following lemma bounds the localized Rademacher complexity in terms of the uniform entropy integral and is a direct consequence of Theorem 2.1 of Van Der Vaart & Wellner (2011).

For any distribution Q and any uniformly bounded function class $\mathcal{F}$, let $N(\varepsilon, \mathcal{F}, L^2(Q))$ denote the ε-covering number of $\mathcal{F}$ under the $L^2(Q)$ norm (Van Der Vaart & Wellner, 1996). Define the uniform entropy integral of $\mathcal{F}$ by

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^{\delta} \sup_{Q} \sqrt{\log N(\epsilon, \mathcal{F}, L^2(Q))}\, d\epsilon,$$

where the supremum is taken over all discrete probability distributions Q.

Lemma 7. *Let $\mathcal{F}$ be a star-shaped class of functions such that $\sup_{f\in\mathcal{F}} \|f\|_\infty \leq M$. Then, for every $\delta > 0$,*

$$\mathcal{R}_n(\mathcal{F}, \delta) \;\lesssim\; \frac{1}{\sqrt{n}}\,\mathcal{J}(\delta, \mathcal{F})\left(1 + \frac{\mathcal{J}(\delta, \mathcal{F})}{\delta\sqrt{n}}\right),$$

where the implicit constant depends only on M.

Proof. This bound follows directly from the argument in the proof of Theorem 2.1 of Van Der Vaart & Wellner (2011); see in particular the step where the local Rademacher complexity is controlled by the uniform entropy integral for star-shaped classes. □

D.2. Regret of a Single Inexact Picard Step

Technical Lemmas. Recall the weight function d_π in Condition C3, and let $\hat{d}_\pi$ denote an estimate of this weight that is trained independently of $\mathcal{D}_n$ (C6).

Let $\mathcal{F}$ be a convex function class and let $\hat{Q}^{\text{init}} \in \mathcal{F}$ be an initial (possibly data-dependent) estimate. Define the stationary-weighted population risk

$$\hat{R}_0(Q) = \mathbb{E}_{\nu_b}\Big[d_\pi(S,A)\,\big\{R + \gamma(\pi\hat{Q}^{\text{init}})(S') - Q(S,A)\big\}^2\Big].$$

Following Algorithm 1, define the empirical minimizer

$$\hat{Q}_n := \arg\min_{Q\in\mathcal{F}} \hat{R}_n(Q),$$

where

$$\hat{R}_n(Q) = \sum_{i=1}^{n} \hat{d}_\pi(S_i, A_i)\,\big\{R_i+\gamma(\pi\hat{Q}^{\text{init}})(S_i')-Q(S_i, A_i)\big\}^2.$$

Lemma 8 shows that this stationary-weighted population objective is correctly specified for the projected Bellman operator $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$ *only* under the orthogonality condition in Condition C3.

Lemma 8 (Fixed-point operator as population risk minimizer)**.** *Under Conditions C3,*

$$\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) = \arg\min_{Q\in\overline{\mathcal{F}}} \hat{R}_0(Q).$$

Proof. Consider the stationary-population objective

$$\tilde{R}_0(Q) := \mathbb{E}_{\mu_\pi}\Big[\big\{R + \gamma(\pi\hat{Q}^{\text{init}})(S') - Q(S,A)\big\}^2 \;\Big|\; \mathcal{D}_n\Big].$$

By definition,

$$\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) = \arg\min_{Q\in\overline{\mathcal{F}}} \tilde{R}_0(Q),$$

as this is precisely the $\mathcal{F}$-projection of the Bellman target $R + \gamma(\pi\hat{Q}^{\text{init}})(S')$.

By the restricted stationary overlap condition in C3, the directional derivatives of $\tilde{R}_0$ and $\hat{R}_0$ coincide. In particular, for every direction $h \in \mathcal{F}$, the first order optiomality conditions for $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$ imply that

$$\begin{aligned}
0 &= \mathbb{E}_{\mu_\pi}\Big[h(S,A)\,\{\mathcal{T}(\hat{Q}^{\text{init}})(S,A) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})(S,A)\}\Big] \\
&= \mathbb{E}_{\mu_\pi}\Big[h(S,A)\,\{R + \gamma(\pi\hat{Q}^{\text{init}})(S') - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})(S,A)\}\Big] \\
&= \mathbb{E}_{\nu_b}\Big[(d_\pi h)(S,A)\,\{R + \gamma(\pi\hat{Q}^{\text{init}})(S') - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})(S,A)\}\Big].
\end{aligned}$$

Thus $\tilde{R}_0$ and $\hat{R}_0$ have identical first-order optimality conditions over $\overline{\mathcal{F}}$, and therefore the same minimizer.

□

The following results control the empirical process fluctuations that arise in the analysis of the inexact Picard iteration. Recall the extended function class

$$\mathcal{F}_{\text{ext}} := \mathcal{F} \cup \mathcal{F}_\pi,$$

where $\mathcal{F}_\pi := \{\,(s,a) \mapsto (\pi f)(s) : f \in \mathcal{F}\,\}$. Recall also the critical radius

$$\delta_n := \inf\Big\{\delta > 0 : \frac{\mathcal{J}(\delta, \mathcal{F}_{\text{ext}})}{\sqrt{n}\,\delta^2} \leq 1\Big\}.$$

Lemma 9. *Let w be a state-action function. Assume that w, R, $\hat{Q}^{\text{init}}$, $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$, and $\hat{Q}_n$ are uniformly bounded by M. Assume further that C6 and C7 hold. Then there exists a constant $C = C(M) > 0$ such that, for all $\eta \in (0,1)$, with probability at least $1-\eta$,*

$$\begin{aligned}&\Big|(P_n - P_0)\big[w\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\,\{R + \gamma(\pi\hat{Q}^{\text{init}}) - \hat{Q}_n\}\big]\Big| \\ &\le C\Big(\delta_n^2 + \delta_n\,\|w(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\|\Big) \\ &\quad + C\Big(\frac{\sqrt{\log(1/\eta)}\,\|w(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n)\|}{\sqrt{n}} + \frac{\log(1/\eta)}{n}\Big),\end{aligned}$$

where $\|\cdot\|$ denotes the $L^2(P_0)$ behavior norm.

Proof. Define the function class

$$\begin{aligned}\mathcal{G} := \Big\{&(s,a,r,s') \mapsto \big(Q_1(s,a) - Q_2(s,a)\big) \\ &\times \big(r + \gamma V_1(s') - Q_2(s,a)\big) : \; Q_1, Q_2 \in \mathcal{F},\; V_1 \in \mathcal{F}_\pi\Big\}.\end{aligned}$$

Without loss of generality, assume that $\mathcal{F}$ and $\mathcal{F}_\pi$ are uniformly bounded by M (for example, by restricting these classes to functions satisfying $\|f\|_\infty \le M$). Since $\mathcal{F}$ and $\mathcal{F}_\pi$ are uniformly bounded, each element of $\mathcal{G}$ can be written as a Lipschitz-continuous pointwise transformation of elements of $\mathcal{F} \times \mathcal{F} \times \mathcal{F}_\pi$, with Lipschitz constant depending only on the uniform bounds and γ. Standard entropy bounds for Lipschitz images of function classes (Van Der Vaart & Wellner, 1996) then imply that there exists a constant $C < \infty$ such that, for all $\delta > 0$,

$$\mathcal{J}(\delta, \mathcal{G}) \lesssim \mathcal{J}(\delta, \mathcal{F}) + \mathcal{J}(\delta, \mathcal{F}_\pi) \lesssim 2\mathcal{J}(\delta, \mathcal{F}_{\text{ext}}).$$

By Lemma 7, the critical radius $\delta_n(\mathcal{G})$ is upper bounded by the uniform entropy-based radius δ_n. Moreover, by C7, we have

$$\frac{\mathcal{J}(\delta, \mathcal{F}_{\text{ext}})}{\delta\sqrt{\log\log(1/\delta)}} \longrightarrow \infty \quad \text{as } \delta \to 0,$$

and hence $n^{-1/2}\sqrt{\log\log(1/\delta_n)} = o(\delta_n)$. Thus, applying Lemma 6, there exists a universal constant $C > 0$ such that, for all $\eta \in (0,1)$, with probability at least $1-\eta$, every $f \in \mathcal{F}$ satisfies

$$\begin{aligned}&\frac{1}{n}\sum_{i=1}^n \big(w(S_i,A_i)f(O_i) - \mathbb{E}[w(S_i,A_i)f(O_i)]\big) \\ &\le C\Big(\delta_n^2 + \delta_n\,\|wf\|\Big) \\ &\quad + C\Big(\frac{\sqrt{\log(1/\eta)}\,\|wf\|}{\sqrt{n}} + \frac{M\,\log(1/\eta)}{n}\Big),\end{aligned}$$

where $O_i = (S_i, A_i, R_i, S_i')$ and $\mathbb{E}_w[f(O)] = \mathbb{E}[w(S,A)f(O)]$. Here we use that the weight function w is uniformly bounded and that multiplication by a fixed bounded function preserves uniform entropy numbers. Thus, the entropy of the class $\{\, wf : f \in \mathcal{G} \,\}$ is controlled by that of $\mathcal{G}$, up to universal constants and the uniform bound.

□

Main bound for inexact Picard iteration. The following lemma bounds the regret incurred by an inexact Picard iteration in terms of the critical radius of $\mathcal{F}$ and the χ^2-divergence of the density ratio d_π. It allows the starting point $\hat{Q}^{\text{init}}$ to be an arbitrary (possibly data-dependent) element of $\mathcal{F}$.

Define the weighted norm:

$$\|f\|_{2,d_\pi}^2 := P_0\big[d_\pi f^2\big].$$

Lemma 10 (Regret for inexact iteration)**.** *Assume that $\hat{d}_\pi$, d_π, $\frac{\hat{d}_\pi}{d_\pi}$, R, $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$, and $\hat{Q}_n$ are uniformly bounded by M. Assume further that C5, C6, and C7 hold. Let $\alpha \in [0,1]$ satisfy C8. Then there exists a constant $C = C(M) > 0$ such that, for all $\eta \in (0,1)$, with probability at least $1-\eta$,*

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\le C\Big(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}} + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,d_\pi}^{\frac{1}{1-\alpha}} \\ &\quad + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_{2,d_\pi} \big\|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\big\|_{2,d_\pi}^{\alpha}\Big).\end{aligned} \tag{4}$$

for some constant $C > 0$. Furthermore, without C8, it holds that

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\le C\Bigg(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}} \\ &\qquad + \Big\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\Big\|_\infty \big\|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\big\|_{2,d_\pi}\Bigg).\end{aligned} \tag{5}$$

for some constant $C > 0$, provided $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_\infty < 1$.

Proof of Lemma 10. By first-order optimality of $\hat{Q}^\star = \arg\min_{Q \in \mathcal{F}} \hat{R}_n(Q)$ and convexity of $\mathcal{F}$, we have, for all $Q \in \mathcal{F}$,

$$\begin{aligned}&\frac{1}{n}\sum_{i=1}^n \hat{d}_\pi(S_i,A_i)\,\{Q(S_i,A_i) - \hat{Q}^\star(S_i,A_i)\} \\ &\qquad \times \{R_i + \gamma\,(\pi\hat{Q}^{\text{init}})(S_i') - \hat{Q}^\star(S_i,A_i)\} \;\le\; 0.\end{aligned}$$

Note that $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) \in \mathcal{F}$ by definition of the projected Bellman operator $\mathcal{T}_{\mathcal{F}}$. Taking $Q = \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n$ in the empirical first-order condition gives

$$\begin{aligned}&\frac{1}{n}\sum_{i=1}^n \hat{d}_\pi(S_i,A_i)\,\{\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})(S_i,A_i) - \hat{Q}_n(S_i,A_i)\} \\ &\qquad \times \{R_i + \gamma\,(\pi\hat{Q}^{\text{init}})(S_i') - \hat{Q}_n(S_i,A_i)\} \le 0.\end{aligned}$$

Adding and subtracting the population expectation and rearranging,

$$P_0\big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{R+\gamma(\pi\hat{Q}^{\text{init}})-\hat{Q}_n\}\big] \le (P_0-P_n)\big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{R+\gamma(\pi\hat{Q}^{\text{init}})-\hat{Q}_n\}\big]. \tag{6}$$

Here we abuse notation and write R for the coordinate projection $(s,a,r,s')\mapsto r$, and similarly view $(\pi\hat{Q}^{\text{init}})$ as the mapping $(s,a,r,s')\mapsto(\pi\hat{Q}^{\text{init}})(s')$.

By the law of total expectations and Adding and subtracting d_π, we obtain

$$\begin{aligned}
&P_0\big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{R+\gamma(\pi\hat{Q}^{\text{init}})-\hat{Q}_n\}\big]\\
&=P_0\big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n\}\big]\\
&=\quad P_0\big[d_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n\}\big]\\
&\qquad+P_0\big[(\hat{d}_\pi-d_\pi)\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n\}\big].
\end{aligned}$$

Applying the law of total expectation,

$$\begin{aligned}
&P_0\Big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{R+\gamma(\pi\hat{Q}^{\text{init}})-\hat{Q}_n\}\Big]\\
&=P_0\Big[d_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,(\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n)\Big]\\
&\quad+P_0\Big[(\hat{d}_\pi-d_\pi)\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n\}\Big].
\end{aligned}$$

The first-order optimality conditions for $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$ as a population minimizer (Lemma 8) imply, for all $\hat{Q}^{\text{init}}\in\mathcal{F}$,

$$P_0\big[d_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{\mathcal{T}(\hat{Q}^{\text{init}})-\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\}\big]\ \ge\ 0.$$

Thus

$$\begin{aligned}
&P_0\Big[d_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,(\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n)\Big]\\
&=P_0\Big[d_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\Big]\\
&\quad+P_0\Big[d_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,(\mathcal{T}(\hat{Q}^{\text{init}})-\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}))\Big]\\
&\ge\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\|_{2,d_\pi}^2.
\end{aligned}$$

By Cauchy–Schwarz,

$$\begin{aligned}
&P_0\big[(\hat{d}_\pi-d_\pi)\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n\}\big]\\
&=P_0\big[d_\pi\big(\tfrac{\hat{d}_\pi}{d_\pi}-1\big)\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n\}\big]\\
&\ge-\Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi}-1\big)(\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n)\Big\|_{2,d_\pi}\big\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\big\|_{2,d_\pi},
\end{aligned}$$

where $\|f\|_{2,d_\pi}^2:=P_0\big[d_\pi f^2\big]$.

Combining the above, we obtain the local quadratic lower bound

$$\begin{aligned}
&P_0\big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{R+\gamma(\pi\hat{Q}^{\text{init}})-\hat{Q}_n\}\big]\\
&\ge\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\|_{2,d_\pi}^2\\
&-\Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi}-1\big)(\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n)\Big\|_{2,d_\pi}\big\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\big\|_{2,d_\pi}.
\end{aligned}$$

Next, applying this lower bound to (6) and rearranging,

$$\begin{aligned}
&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\|_{2,d_\pi}^2\\
&\le(P_0-P_n)\big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{R+\gamma(\pi\hat{Q}^{\text{init}})-\hat{Q}_n\}\big]\\
&\quad+\Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi}-1\big)(\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n)\Big\|_{2,d_\pi}\big\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\big\|_{2,d_\pi}.
\end{aligned}\tag{7}$$

Denoting the Bellman residual by

$$\xi_n:=R+\gamma(\pi\hat{Q}^{\text{init}})-\hat{Q}_n,$$

the empirical process term satisfies

$$\begin{aligned}
&(P_0-P_n)\Big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\xi_n\Big]\\
&=(P_0-P_n)\Big[d_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\xi_n\Big]\\
&\quad+(P_0-P_n)\Big[(\hat{d}_\pi-d_\pi)\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\xi_n\Big].
\end{aligned}$$

We now invoke C6 and apply Lemma 9 conditionally on the data used to train the weights to the two terms on the right-hand side, with $w=d_\pi$ and $w=\hat{d}_\pi-d_\pi$, both of which are uniformly bounded by Assumption C4. This yields a constant $C>0$ such that, for all $\eta\in(0,1)$, with probability at least $1-\eta$,

$$\begin{aligned}
&\Big|(P_0-P_n)\big[\hat{d}_\pi\,(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\,\{R+\gamma(\pi\hat{Q}^{\text{init}})-\hat{Q}_n\}\big]\Big|\\
&\le C\Big(\delta_n^2+\delta_n\,\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\|_{2,d_\pi}\Big)\\
&+\ C\left(\frac{\sqrt{\log(1/\eta)}\,\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\|_{2,d_\pi}}{\sqrt{n}}+\frac{\log(1/\eta)}{n}\right).
\end{aligned}$$

Combining this high-probability bound with (7), we conclude

$$\begin{aligned}
&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\|_{2,d_\pi}^2\\
&\le C\Big(\delta_n^2+\delta_n\,\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\|_{2,d_\pi}\Big)\\
&+\ C\left(\frac{\sqrt{\log(1/\eta)}\,\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\|_{2,d_\pi}}{\sqrt{n}}+\frac{\log(1/\eta)}{n}\right)\\
&+\ \Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi}-1\big)(\mathcal{T}(\hat{Q}^{\text{init}})-\hat{Q}_n)\Big\|_{2,d_\pi}\big\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\big\|_{2,d_\pi}.
\end{aligned}\tag{8}$$

The inequality above, together with the loose bound

$$\begin{aligned}
&\Big\|\big(\tfrac{\hat{d}_\pi}{d_\pi}-1\big)(\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n)\Big\|_{2,d_\pi}\\
&\lesssim\Big\|\tfrac{\hat{d}_\pi}{d_\pi}-1\Big\|_{2,d_\pi}
\end{aligned}\tag{9}$$

implies that, with probability at least $1-\eta$,

$$\begin{aligned}
&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})-\hat{Q}_n\|_{2,d_\pi}\\
&\qquad\le C\left(\delta_n+\Big\|\tfrac{\hat{d}_\pi}{d_\pi}-1\Big\|_{2,d_\pi}+\sqrt{\frac{\log(1/\eta)}{n}}\right),
\end{aligned}\tag{10}$$

for some constant C.

To obtain a sharper bound, we invoke Condition C8: there exists some $\alpha \in [0,1]$ such that

$$\|\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_\infty \lesssim \|\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}^\alpha.$$

Then, by Hölder's inequality, the bound in (9) can be improved to

$$\begin{aligned}&\left\|\left(\tfrac{\hat{d}_\pi}{d_\pi} - 1\right)\left(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\right\|_{2,d_\pi} \\ &\lesssim \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_{2,d_\pi} \times \|\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi}^\alpha.\end{aligned}$$

We can then apply Lemma 9 together with (7) and argue as before to conclude that, with probability at least $1-\eta$,

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\le C\Big(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}} + \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_{2,d_\pi}^{\frac{1}{1-\alpha}} \\ &\quad + \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_{2,d_\pi} \left\|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\right\|_{2,d_\pi}^\alpha\Big).\end{aligned} \tag{11}$$

for some constant $C > 0$. The claim then follows, up to a change in the constant, by invoking Condition C5 to replace the weighted norm with the $L^2(\mu_\pi)$ norm.

Bound with sup-norm weight error. If one is willing to measure weight accuracy in supremum norm, an alternative bound for (9) is

$$\begin{aligned}&\left\|\left(\tfrac{\hat{d}_\pi}{d_\pi} - 1\right)\left(\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\right)\right\|_{2,d_\pi} \\ &\le \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_\infty \|\mathcal{T}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\le \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_\infty \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\quad + \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_\infty \|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\|_{2,d_\pi}.\end{aligned}$$

In that case, we obtain

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) - \hat{Q}_n\|_{2,d_\pi} \\ &\le C\Bigg(\delta_n + \sqrt{\tfrac{\log(1/\eta)}{n}} \\ &\qquad + \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_\infty \left\|\mathcal{T}(\hat{Q}^{\text{init}}) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})\right\|_{2,d_\pi}\Bigg).\end{aligned} \tag{12}$$

for some constant $C > 0$, provided $\|\frac{\hat{d}_\pi}{d_\pi} - 1\|_\infty < 1$.

□

D.3. Main result

Proof of Lemma 3. Under the stated conditions, the claim follows directly from Lemma 10. □

Proof of Theorem 2. By Lemma 3 and a union bound, there exists a constant $C = C(M) > 0$ such that, for all $\eta \in (0,1)$ and all $k \in [K]$, with probability at least $1-\eta$,

$$\begin{aligned}\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)}) - \hat{Q}^{(k)}\|_{2,\mu_\pi} &\le C\left(\delta_n + \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_{2,\mu_\pi}\right) \\ &\quad + C\sqrt{\frac{\log(K/\eta)}{n}}.\end{aligned}$$

Define, for each k,

$$\varepsilon_k := C\left(\delta_n + \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_{2,\mu_\pi} + \sqrt{\frac{\log(K/\eta)}{n}}\right),$$

so that on this event,

$$\|\hat{Q}^{(k)} - \mathcal{T}_{\mathcal{F}}(\hat{Q}^{(k-1)})\|_{2,\mu_\pi} \le \varepsilon_k \qquad \text{for all } k \in [K].$$

Thus, by Lemma 2, for any $K \ge 1$,

$$\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \le \gamma^K \|\hat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} + \sum_{j=1}^K \gamma^{K-j}\varepsilon_j.$$

Since ε_j does not depend on j, write $\varepsilon_j = \varepsilon$ for all $j \in [K]$. Hence,

$$\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \le \gamma^K \|\hat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} + \varepsilon\,\frac{1-\gamma^K}{1-\gamma}.$$

Thus,

$$\begin{aligned}\|\hat{Q}^{(K)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} &\le \gamma^K \|\hat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu_\pi} \\ &\quad + \frac{C}{1-\gamma}\left(\delta_n + \left\|\tfrac{\hat{d}_\pi}{d_\pi} - 1\right\|_{2,\mu_\pi}\right) \\ &\quad + \frac{C}{1-\gamma}\left(\sqrt{\frac{\log(K/\eta)}{n}}\right).\end{aligned}$$

□

E. Known transitions and estimated rewards

Proof of Theorem 3. By Lemma 5 and stationarity of μ_π, $\mathcal{T}_{\mathcal{F}}$ is a γ-contraction on $L^2(\mu_\pi)$.

By definition of the fixed points,

$$Q^\star = \mathcal{T}_{\mathcal{F}}(Q^\star), \qquad \hat{Q}^\star = \hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star).$$

Write

$$\hat{Q}^\star - Q^\star = \hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(Q^\star),$$

and add and subtract $\mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)$:

$$\hat{Q}^\star - Q^\star = \underbrace{\left(\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)\right)}_{(I)} + \underbrace{\left(\mathcal{T}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(Q^\star)\right)}_{(II)}.$$

Taking $L^2(\mu_\pi)$-norms and using the triangle inequality,

$$\|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi} \le \|\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)\|_{2,\mu_\pi} + \|\mathcal{T}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(Q^\star)\|_{2,\mu_\pi}.$$

The second term is controlled by the contraction property:

$$\|\mathcal{T}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(Q^\star)\|_{2,\mu_\pi} \le \gamma \|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi}.$$

Hence

$$(1-\gamma)\|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi} \le \|\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)\|_{2,\mu_\pi}. \quad (13)$$

Next, observe that for any Q,

$$\begin{aligned}\|\hat{\mathcal{T}}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q)\|_{2,\mu_\pi} &= \|\Pi_{\mathcal{F}}\hat{\mathcal{T}}(Q) - \Pi_{\mathcal{F}}\mathcal{T}(Q)\|_{2,\mu_\pi} \\ &\le \|\hat{\mathcal{T}}(Q) - \mathcal{T}(Q)\|_{2,\mu_\pi} \\ &= \|\hat{r} - r\|_{2,\mu_\pi},\end{aligned}$$

since $\hat{\mathcal{T}}(Q) - \mathcal{T}(Q) = \hat{r} - r$ does not depend on Q. Substituting into (13) yields

$$\|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi} \le \frac{1}{1-\gamma}\|\hat{r} - r\|_{2,\mu_\pi}.$$

Stronger guarantee under linear function classes. If $\mathcal{F}$ is a linear subspace of $L^2(\mu_\pi)$, then $\Pi_{\mathcal{F}}$ is a linear orthogonal projection, and we obtain a sharper reward misspecification bound. In this case, for any Q,

$$\begin{aligned}\|\hat{\mathcal{T}}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q)\|_{2,\mu_\pi} &= \|\Pi_{\mathcal{F}}\hat{\mathcal{T}}(Q) - \Pi_{\mathcal{F}}\mathcal{T}(Q)\|_{2,\mu_\pi} \\ &= \|\Pi_{\mathcal{F}}\big(\hat{\mathcal{T}}(Q) - \mathcal{T}(Q)\big)\|_{2,\mu_\pi} \\ &= \|\Pi_{\mathcal{F}}(\hat{r} - r)\|_{2,\mu_\pi},\end{aligned}$$

since $\hat{\mathcal{T}}(Q) - \mathcal{T}(Q) = \hat{r} - r$ does not depend on Q. In particular,

$$\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star) = \Pi_{\mathcal{F}}(\hat{r} - r),$$

so

$$\|\hat{\mathcal{T}}_{\mathcal{F}}(\hat{Q}^\star) - \mathcal{T}_{\mathcal{F}}(\hat{Q}^\star)\|_{2,\mu_\pi} = \|\Pi_{\mathcal{F}}(\hat{r} - r)\|_{2,\mu_\pi}.$$

Substituting this into (13) yields

$$\|\hat{Q}^\star - Q^\star\|_{2,\mu_\pi} \le \frac{1}{1-\gamma}\|\Pi_{\mathcal{F}}(\hat{r} - r)\|_{2,\mu_\pi},$$

so only the component of the reward error lying in $\mathcal{F}$ affects the fixed point.

□